

Cerberus — Submission Brief

3-Agent Autonomous Security on Somnia Agentic L1

Three AI agents. One 60-second heartbeat. Zero trust required.

What Is Cerberus?

Cerberus is an autonomous security network built on Somnia's Agentic L1. Three specialized AI agents — OracleGuard, ThreatClassifier, and CircuitBreaker — run a continuous 60-second pipeline that detects oracle anomalies, classifies threats via deterministic AI, and automatically pauses compromised contracts. Every decision produces a verifiable on-chain receipt.

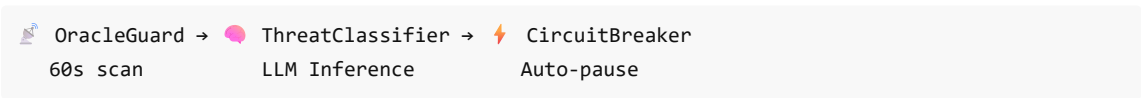
It is not a monitoring tool or a static scanner. It is an immune system for smart contracts — always watching, always reasoning, always ready to act.

The Problem It Solves

Smart contracts have no autonomous defense mechanism. When a Chainlink oracle is manipulated, a price feed diverges from consensus, or an exploit pattern emerges, no one detects it in real time. By the time a human notices and responds — funds are already gone. The 2023 Euler Finance hack drained \$197 million in minutes.

Existing solutions are either reactive (post-exploit forensic analysis), centralized (a single monitoring bot that can be DDoSed or bribed), or rule-based (hardcoded thresholds that miss novel attacks). Cerberus replaces all three with autonomous AI agents that watch, reason, and protect — natively on Somnia.

The Pipeline



OracleGuard fetches ETH/USD from 3 independent sources (CoinGecko, Binance, CryptoCompare). When any feed diverges >2%, ThreatClassifier uses Somnia's deterministic LLM to classify the threat. If CRITICAL, CircuitBreaker pauses the protected contract on-chain — embedding a verifiable receipt for permanent audit.

Somnia Primitives Used

Primitive	Agent	Purpose
JSON API Request	OracleGuard	Fetch live price data from 3 external APIs
LLM Inference	ThreatClassifier	Deterministic AI threat classification with consensus-verified outputs
Verifiable Receipts	All 3 agents	Every anomaly, classification, and pause proven on-chain

On-Chain Proof

Agent Wallet: 0x94A4365E6B7E79791258A3Fa071824BC2b75a394

Sentinel Contract: 0x87E3D9fcfA4eff229A65d045A7C741E49b581187

Chain: Somnia Testnet (50312)

Live Transactions:

Action	Tx Hash
protect() — Register contract for monitoring	0xd22a8de50626b7902b4fed3879213e50ea41572a5f85105677d3d11fed0b4655
unpauseContract() — Reset for fresh demo	0x76ed43595135667fa3943b86ee68450a9eaa529f9a70e23e543c6e9bcb59e5de

Judging Criteria

Functionality: Deployed on Somnia Testnet. 23+ pipeline iterations without critical failure. 3 live price sources. Contract verified on-chain.

Agent-First Design: All 3 agents operate natively on Somnia. JSON API Request for data fetching. LLM Inference for deterministic classification. Agents discover anomalies, invoke each other, and act autonomously.

Innovation: First on-chain security sentinel using deterministic LLM for threat classification. Not rule-based — AI reasoning with verifiable cryptographic receipts.

Autonomous Performance: 60-second self-healing heartbeat. Error-isolated pipeline. Survives API failures and RPC drops. Runs indefinitely.

What Makes It Different

Most security tools are reactive (scan after deployment) or rule-based (hardcoded thresholds). Cerberus uses deterministic AI to reason about threats in real time, with every decision producing a verifiable receipt proving the AI caught it autonomously. It is not a scanner — it is an immune system.

Traditional monitoring relies on a single bot that can be DDoSed, bribed, or coerced. Cerberus distributes detection, classification, and response across three specialized agents — each running independently, each with verifiable outputs. No single point of failure. No single point of trust.

Links

- **GitHub:** <https://github.com/Gideon145/cerberus>
- **Live Demo:** <https://cerberus-production-8429.up.railway.app>
- **Demo Video:** https://youtube.com/shorts/oDqEUR_oynA
- **Presentation:** <https://cerberus-production-8429.up.railway.app/presentation>

Stack

Solidity 0.8.20 · TypeScript (Node.js) · Somnia JSON API Request · Somnia LLM Inference · ethers.js v6 · Railway · Single-file HTML dashboard

"Smart contracts have no immune system. Cerberus is that immune system."